

Benjamin Rémi Berton

Human Factors & Cognitive Engineering Researcher
PhD Candidate · Human–Autonomy Teaming in Aviation

PROFILE

Human factors researcher and engineer designing and evaluating autonomous flight-deck systems. I combine coactive design, cognitive modelling, agent prototyping, and human-in-the-loop simulation to evaluate function allocation in terms of cognitive workload, situation awareness, trust, and pilot oversight, with the goal of building safer human–autonomy concepts in aviation.

EXPERIENCE

Doctoral Researcher · Polytechnique Montréal

Sep 2023–Present

Department of Mathematics and Industrial Engineering · Doctoral research funded through a CRIAQ collaborative project involving McGill University, Polytechnique Montréal, Ryerson University, Bombardier, Thales, Marinvent, IATA, Adacel, CMC Microsystems and NAV CANADA.

- Built a modular closed-loop testbed connecting a cognitive pilot model, autonomous cockpit agent and interface, and flight simulator (X-Plane) through the local-area network to compare teaming designs before costly pilot studies.
- Derived function-allocation patterns with Interdependence Analysis and evaluated four agent configurations in a human-in-the-loop study with 18 professional pilots, combining flight performance, workload, trust, oversight behaviour, and interview data.
- Modelled verification, dynamic trust calibration, reactive reliance, and visual attention within a cognitive architecture (QN-ACTR/SEEV); developed team-, task-, interaction-, and cognition-level simulation measures.

Course Instructor & Teaching Assistant · Polytechnique Montréal

2024–2026

Graduate teaching · Department of Mathematics and Industrial Engineering

- Instructor for IND8412A — *User Experience for the Web* in Winter 2024, 2025, and 2026; nominated for a Best Teacher Award following the first course delivery. Teaching assistant for IND6406 — *Cognitive Ergonomics* in Winter 2026.

Human Factors Researcher · ESTIA

Sep 2022–Sep 2023

PEPSS platform · Airbus/CORAC TOUCANS project · Bidart, France

- Developed a practitioner-oriented methodology to assess physical, physiological, and cognitive effects of head-mounted augmented reality in constrained environments; contributed to a peer-reviewed conference paper.
- Teaching assistant for Web Programming and Introduction to Ergonomics.

Prototype Development Engineer / Human Factors Specialist

Jan–Sep 2022

MBDA · Human-centred training prototypes

Embedded Systems Developer Intern

Apr–Jul 2021

TELERAD · ACARS/VDL2 integration for 9000-2G transceivers

Web Development Internships

2019–2020

Customeyes & 100% Digital · AR/WebRTC prototypes

SELECTED PUBLICATIONS

Berton, B. R., & Doyon-Poulin, P. (2026). *A Cognitive Modelling Framework for Simulating Human–Autonomy Teaming in the Flight Deck*. HCI International, 333–352.

Berton, B., Bouzekri, E., Singh, A., & Doyon-Poulin, P. (2026). *Human–Autonomy Teaming for Future Commercial Aviation Concepts of Operations*. Handbook of Sociotechnical Systems, 428–446.

Bouzekri, E., Berton, B., & Doyon-Poulin, P. (2025). *Who Is Responsible? Roles of Future Pilots in Single-Pilot Operations*. Human Work Interaction Design.

Berton, B. R., Morel-Targosz, C., Chaumillon, R., & Wolff, M. (2024). *Paving the Way Towards a Methodology to Assess the Impacts of Augmented Reality Under Constrained Environments: A Head-Mounted Display Use Case*.

SELECTED OPEN-SOURCE SOFTWARE

Exoplanet Explorer (C# - GODOT) — MIT-licensed educational game and HAT testbed used by ≈ 100 graduate students; archived at [doi:10.5281/zenodo.18651448](https://doi.org/10.5281/zenodo.18651448).

Interdependence Analysis Dashboard (Python / Plotly Dash) — Interactive editor and visual analytics for team capabilities, task allocations, and teaming requirements; available on [GitHub](#).

TARS Cockpit Agent (Python / Qt) — Finite-state-machine autonomous agent with a PySide Qt interface, Silero text-to-speech, and Vosk speech-to-text.

QuestLab (Python / Flask / SQLite) — Local-network platform for configurable questionnaire batteries in laboratory experiments.

CONTACT

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EXPERTISE

Human–autonomy teaming

Cognitive engineering

Aviation human factors

Systems Engineering

Situation awareness

Cognitive workload

Function allocation

Human-Centered Design

METHODS

Cognitive architecture (ACT-R)

Human-performance modelling (SEEV)

Interdependence Analysis

Task analysis

Human-in-the-loop simulation

Experiment design

Mixed-method evaluation

Eye tracking · EMG · HRV · ECG · EDA

TECHNICAL SKILLS

Python · Java · C#

Common lisp · Web (JS,HTML,CSS)

Qt/QML · Figma · Git

EDUCATION

PhD · Industrial Engineering

Polytechnique Montréal

2023–Present

Supervisor: Philippe Doyon-Poulin

M. Sc · Ergonomics & Human Factors Engineering

Université Paris Cité

2021–2022 · With honours

Research focus: neuroergonomics

M. Eng. · Digital Design & Innovation

ESTIA Institute of Technology

2020–2022

BTech · Engineering

ESTIA Institute of Technology · 2017–2020

Academic exchange in Computer Engineering · East China University of Science and Technology · 2019–2020

LANGUAGES

French

English · IELTS 7.5 / TOEIC 925

Spanish · B1

Mandarin Chinese · HSK 3

ADDITIONAL TRAINING

French Navy preparation

French Sailing Pilot Licence

Brevet d'initiation à l'aéronautique (BIA)

Car & motorcycle driving licences

PSC1 first-aid certification